

CS 3307 Project Proposal

Single Player Rhythm Game Engine

Group Number: 75

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1 Problem Statement

Rhythm games are a genre of the game where players need to respond to musical patterns by performing inputs at the correct time. Although the basic interaction (pressing a key when a note reaches a target) is simple, the game must coordinate timing, note patterns, scoring, combo tracking, difficulty, progression, and persistent player data. Small errors in these systems can produce inconsistent gameplay, such as notes being judged differently depending on timing, scores being calculated incorrectly, or player progress being lost. The proposed system is a desktop rhythm game that provides a structured environment for playing songs, receiving timing-based judgments, and tracking progression.

The primary users are players, who select songs and difficulties, play through the charts, and view their results. A player should be able to understand how well they performed through information such as score, accuracy, judgement counts, and combo.

A secondary stakeholder is the developer, who needs the game to have well-structured domain model so that scoring rules, difficulty configurations, and other gameplay variations can be added without substantially modifying the existing components. The project also needs to be maintainable and testable so that changes to one part of the system do not unexpectedly break other parts.

The main difficulty is that a rhythm game involves several components that must cooperate while remaining independent. For instance, gameplay depends on precise timing, but timing should not be tied directly to user interface or a particular timing implementation. Similarly, judging a note, calculating a score, maintaining a combo, and determining progression are different responsibilities that should not be concentrated in one large game class.

2 Requirements

The initial requirements were gathered from three sources. First, the project's required Design Depth Profile and technical requirements were used to identify the architectural and domain capabilities that the system must support.

Second, multiple rhythm games that already exist were examined to identify common gameplay concepts such as song selection, note charts, timing judgements, combos, scoring, and results. Examples considered include well-known rhythm games such as *osu!mania* and *DJMAX RESPECT V*.

Finally, the proposed gameplay was considered from the perspective of a typical player to identify the actions that the user must be able to perform.

2.1 Function Requirements

2.1.1 FR1

Display a list of available songs and select a song to play. Must have.

2.1.2 FR2

Select an available difficulty for a selected song. Must have.

2.1.3 FR3

Load the selected song's chart and present its notes to the player during gameplay. Must have.

2.1.4 FR4

Accept player input for a note and determine a timing judgement based on difference between the player's input time and the note's scheduled (intended) time. Must have.

2.1.5 FR5

Support 2 types of notes: tap, and hold notes. Each type must apply its appropriate gameplay behaviour when played. Must have.

2.1.6 FR6

Calculate and display the player's score, combo, and accuracy during or after a song based on player's note judgements. Must have.

2.1.7 FR7

Allow player to pause, resume, and complete an active game session, while rejecting invalid game states. Must have.

2.1.8 FR8

Save player's progression and song results and restore them everytime the application is opened. Must have.

2.1.9 FR9

Different game modes using different controls. Such as keyboard-based modes, and game mode that you control only with the mouse. Could have.

2.1.10 FR10

Unlock songs or difficulties when the player's progression satisfies the unlock conditions such as, but not limited to, achieve certain threshold of accuracy, or play a song certain times. Note that this will require to modify FR2 to add functionality that prevent player to choose locked songs/difficulties depending on their progression. Should have.

2.1.11 FR11

The system shall display an appropriate error message and preserve the existing valid state when a save file is missing or corrupted. Must have.

2.2 Non-functional Requirements

2.2.1 NFR1 - Performance

The system shall process a valid gameplay input and produce its corresponding domain judgement without noticeable delay during normal gameplay. Under normal single player gameplay conditions, the domain operation for processing a note input should complete within 50 ms.

2.2.2 NFR2 - Reliability

The system shall not terminate/crash unexpectedly as a result of invalid player input, illegal game state transitions, or a corrupt save file. These conditions shall produce a defined error messages instead.

2.2.3 NFR3 - Maintainability

The domain logic shall remain independent of Qt user interface classes. Domain classes shall be compilable and testable without the user interface part of the application.

2.2.4 NFR4 - Usability

A player shall be able to start a song from the selection screen using the available controls without needing to configure files or interact with the application's underlying data files.

3 Use Case Model

3.1 Use Cases

3.1.1 UC1 - Select Song

The player views the available songs and selects a song to play.

3.1.2 UC2 - Select Difficulty

The player selects an available difficulty for the chosen song.

3.1.3 UC3 - Select Game Mode

The player selects a supported game mode and its associated input controls.

3.1.4 UC4 - Play Song

The player provides inputs while notes appear, and the system judges the inputs and updates the player's performance.

3.1.5 UC5 - Pause/Resume Game

The player pauses or resumes an active game session.

3.1.6 UC6 - Complete Song

The system detects the end of a song, calculates the final performance, and records the song result.

3.1.7 UC7 - View Result

The player views their final score, combo, accuracy, and note judgements after completing a song.

3.1.8 UC8 - Save Progress

The system saves the player's progression and song results to persistent storage.

3.1.9 UC9 - Load Progress

The system restores the player's saved progression and results when the application starts.

3.1.10 UC10 - Unlock Content

The system evaluates the player's progression and unlocks songs or difficulties when their unlock conditions are satisfied.

3.1.11 UC11 - Handle Invalid Save

The system detects a missing or corrupted save file, informs the player, and preserves or creates a valid profile state.

3.2 Traceability Table

Functional Requirements	Realized Use Case(s)
FR1	UC1
FR2	UC2, UC10
FR3	UC1, UC4
FR4	UC4
FR5	UC4
FR6	UC4, UC6, UC7
FR7	UC4, UC5, UC6
FR8	UC8, UC9
FR9	UC3, UC4
FR10	UC2, UC10
FR11	UC9, UC11

Table 1: Traceability Table

3.3 UML Use Case Diagram

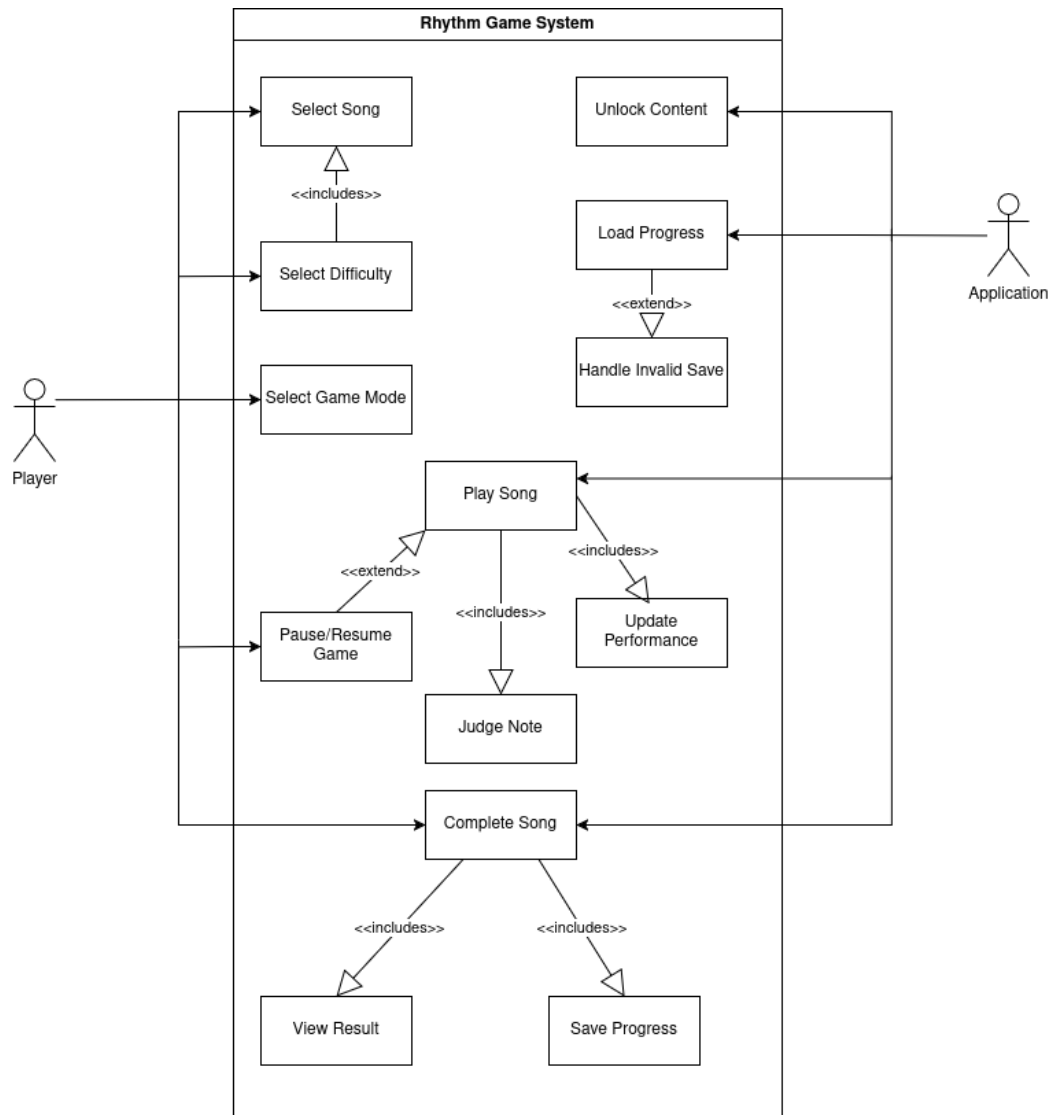


Figure 1: Use Case Diagram

3.4 Written Use Case - UC4

3.4.1 Primary Actor

Player

3.4.2 Goal

Play a selected song and receive timing judgements and performance feedback based on their inputs.

3.4.3 Preconditions

1. The application is running.
2. The player has selected a valid song.
3. The player has selected an available difficulty.
4. The selected chart can be loaded successfully.
5. A valid game mode has been selected.
6. A 'GameSession' can be created in the initial state.

3.4.4 Main Success Scenario

1. The player selects Play from the song selection screen.
2. The system loads the selected chart.
3. The system creates a new game session using selected song, difficulty, and game mode.
4. The game session transitions from 'Loading' to 'Playing'.
5. The system begins presenting notes according to their scheduled times.
6. The player provides an input corresponding to an upcoming note.
7. The system obtains the input timestamp.
8. The system identifies the relevant note.
9. The system calculates the difference between the player's input time and the note's scheduled time.

10. The system determines a timing judgement, such as 'Perfect', 'Great', 'Good', or 'Miss'.
11. The system applies the appropriate behaviour for the note type. For example, a tap note is resolved by a single input, while a hold note requires the player to maintain the input for its required duration.
12. The system updates the player's score, combo, and accuracy.
13. The updated performance information is displayed to the player.
14. Steps 6-13 repeat as the remaining notes are played.
15. When the chart is complete, the system transitions the game session to 'Finished'.
16. The system creates a song result containing the player's score, accuracy, combo, and judgement information.
17. The result is displayed to the player.
18. The player's progression and result are saved.

3.4.5 Alternative Flows

- A1: Player pauses the game.
 1. While the session is 'Playing', the player selects pause.
 2. The system transitions the session to 'Paused'.
 3. Note progression and gameplay input processing stop.
 4. The player selects resume.
 5. The system transitions the session back to 'Playing'.
 6. Gameplay continues from the paused state.
- A2: Player completes a hold note.
 1. The player begins the hold input within the valid timing window.
 2. The system records the start of the hold.
 3. The player maintains the input for the required duration.
 4. The system validates the hold duration.
 5. The hold note is successfully resolved and the appropriate score is awarded.

- A3: Player makes a late or early input.
 1. The player's input occurs outside the 'Perfect' timing window.
 2. The system calculates the timing difference.
 3. The appropriate lower judgement is assigned.
 4. The score, combo, and accuracy are updated according to the judgement.

3.4.6 Exception Flows

- E1: Invalid player input
 1. The player provides an input that does not correspond to a valid control or lane.
 2. The system rejects the input.
 3. The system displays an appropriate feedback message where applicable.
 4. The current game state, score, and valid notes remain unchanged.
- E2: Illegal game state transition
 1. An attempt is made to perform an operation that is invalid for the current state, such as resuming a session that is already 'Playing'.
 2. The system rejects the transition.
 3. The session remains in its current valid state.
 4. The system reports the error appropriately.
- E3: Chart cannot be loaded
 1. The selected chart cannot be loaded or is invalid.
 2. The system does not start the game session.
 3. The player is informed that the song cannot currently be played.
 4. The player remains on the song selection screen.

3.4.7 Postconditions

- Successful Completion
 1. The game session is in the 'Finished' state.

2. Every required note has been resolved.
 3. A song result containing the player's performance has been created.
 4. Score, combo, accuracy, and judgement information have been calculated.
 5. The result is displayed to the player.
 6. The player's progression is updated.
 7. The updated progression is saved.
- Unsuccessful Completion
 1. The game session remains in a valid state or is not created.
 2. No invalid score or progression data is stored.
 3. The player receives appropriate feedback about the failure.

4 Initial Class Diagram

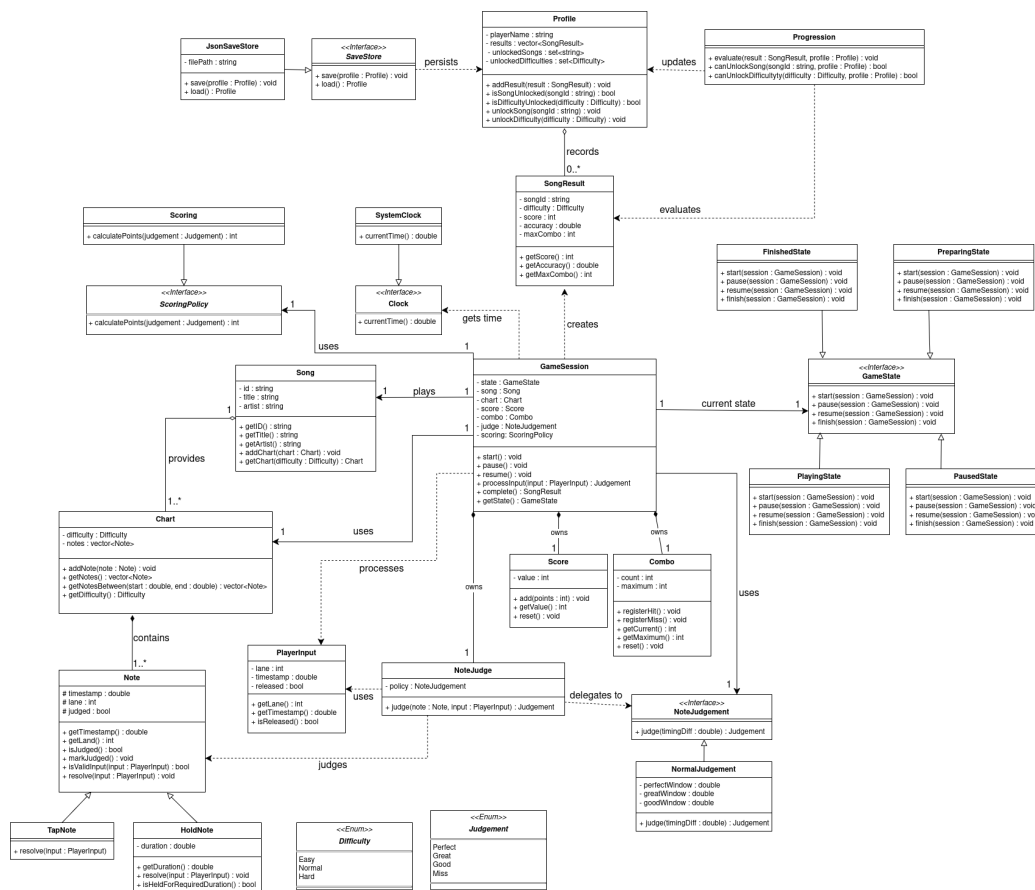


Figure 2: Initial Class Diagram

Since the diagram is big, you may not clearly see everything on this document.

If that's the case, see [image on github](#) for details.

5 Sequence Diagram

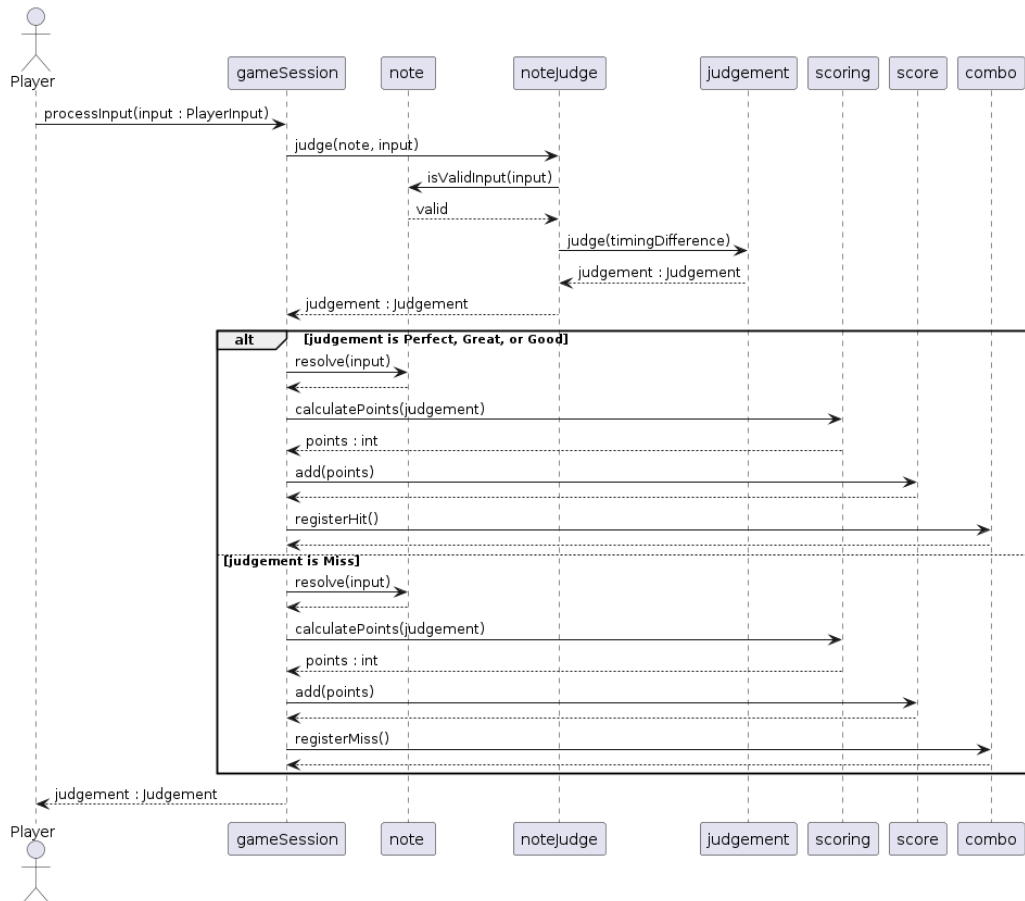


Figure 3: Sequence Diagram

If you can't read some texts, see [image on github](#).

6 Architecture Overview

The system will use a modular, layered architecture consisting of three core domain modules: Song and Chart, Gameplay, and Progression. The system will also contain Qt-based presentation layer in `ui/` and an external service boundary within `core/` for persistence and time-related dependencies.

The Song and Chart module is responsible for representing the available songs, difficulty-based charts, and individual notes. A Song contains one or more Chart objects, while Chart contains Note objects. Note is an abstract class with TapNote and HoldNote implementations. This module contains the data and behaviour required to describe what the player must perform, but does not handle user interface concerns or the overall game session.

The Gameplay module controls an active game session. GameSession coordinates the current song, chart, score, combo, note judgement, scoring policy, and game state transitions. NoteJudge determines the result of player input using NoteJudgement abstraction, while ScoringPolicy determines the points awarded for a judgement. The GameState and its states enforce valid game session transitions. The Gameplay module depends on the Song and Chart module because it processes the notes contained in a selected chart.

The Progression module manages information that persists between gameplay sessions. Profile stores the player's results and unlocked content, SongResult represents the outcome of a completed song, and Progression evaluates results to determine whether songs or difficulties should be unlocked. The Progression module therefore receives results from Gameplay and maintains the player's progression.

The UI layer is located in `ui/` and forms a strict boundary around the core domain. It will contain the Qt 6 views and controllers used to display menus, song selection, gameplay, and results. UI classes may depend on classes and interfaces in `core/`, but the reverse dependency is prohibited. For example, the UI may translate a keyboard event into a PlayerInput and pass it to GameSession, but GameSession will not depend on Qt classes. This keeps the domain independently compilable and testable.

The external boundary is handled through two interfaces in `core/`: SaveStore and Clock. SaveStore abstracts persistence so that the domain does not depend directly on a particular file format. Its real implementation will be JsonSaveStore, which saves and loads the player's Profile from a file. Similarly, Clock abstracts access to the current time. SystemClock will provide the real implementation during gameplay. Consequently, the core domain depends on these abstractions rather than directly on external services.

7 Design Approach & Candidate Patterns

Encapsulation will be used to keep domain state private and expose only operations that maintain valid state. For example, Score will control changes to its score value through operations such as `add()`, while GameSession will control valid transitions between its game states rather than allowing other classes to directly modify its state.

Inheritance will be used where there is genuine is-a relationship and shared behaviour. TapNote and HoldNote will inherit from the abstract Note class because both represent playable notes but have different resolution behaviour.

Polymorphism will allow GameSession and other domain classes to work with abstractions rather than concrete implementations. For example, NoteJudge uses the NoteJudgement interface, allowing a NormalJudgement implementation to be replaced by another judgement policy without modifying NoteJudge. GameSession similarly uses the ScoringPolicy interface rather than depending directly on Scoring. This supports adding alternative game-play rules without modifying existing classes.

Abstraction will be important at variation points and external boundaries. NoteJudgement, ScoringPolicy, GameState, Clock, and SaveStore expose only the operations that clients need while hiding their implementation details. The Clock and SaveStore abstractions also ensure that the core domain does not depend directly on the system.

Where possible, the design will favour composition over inheritance. For example, GameSession contains Score, Combo, and NoteJudge rather than inheriting from these classes. Composition allows each responsibility to change independently and avoids creating a deep inheritance hierarchy. Inheritance is only reserved for cases such as TapNote/HoldNote mentioned above.

The design will be primarily influenced by the SOLID principles:

- **Single Responsibility:** classes such as NoteJudge, Score, Combo, Progression, and GameSession will have distinct responsibilities rather than combining unrelated gameplay logic.
- **Open/Closed:** judgement and scoring behaviour can be extended by adding implementations of NoteJudgement and ScoringPolicy without modifying their clients.
- **Liskov Substitution:** concrete notes, judgement policies, and game states should be usable wherever their corresponding abstractions are expected.

- **Interface Segregation:** interfaces such as Clock and SaveStore will expose only the operations needed by their clients.
- **Dependency Inversion:** GameSession and other domain classes will depend on abstractions such as Clock, SaveStore, NoteJudgement, and ScoringPolicy rather than concrete implementations.

The state pattern is currently the strongest pattern design candidate because the game session has a clearly defined lifecycle with guarded transitions. This choice will be revisited during Deliverable 2 as the implementation and class responsibilities become clearer.

8 Milestone Plans & Risks

D2 will provide about half of the final system, with priority given to establishing the complete core architecture and a working gameplay loop. The remaining features will then build on this foundation rather than requiring major architectural changes late in the project.

Requirements	D2 Target	Planned implementation
FR1	Yes	Song selection screen with available songs
FR2	Yes	Difficulty selection for selected song
FR3	Partial	Chart loading logic and draw notes on screen
FR4	Partial	NoteJudge timing window
FR5	Yes	TapNote and HoldNote implementation
FR6	Partial	Score and combo calculation
FR7	Partial	Core GameState lifecycle, and UI integration for main transitions
FR8	Partial	SaveStore abstraction, JsonSaveStore logic, and basic profile/result persistence
FR9	No	Optional feature deferred until all FR is complete
FR10	No	Planned for D3
FR11	Partial	Persistence error detection and failure behaviour

Table 2: Milestone Planning Table

The first risk is that gameplay architecture getting too complex. This could potentially require major modification to notes, scoring, judgement, or game states and make D2/D3 implementation difficult to maintain. Possible mitigation is to implement a small end-to-end gameplay slice early. Keep `GameSession` as a coordinator and separate judgement, scoring, notes, state management, and persistence behind focused classes and interfaces.

The second risk is that timing and input handling being difficult to implement reliably. In rhythm games, timing is the key. And inaccurate timing could make the core system mechanic unreliable and difficult to test. Possible mitigation is to separate timing behind the `Clock` interface and represent input using `PlayerInput` class. By creating second `Clock` interface, we can use that to create deterministic tests for timing windows and note judgements rather than relying on real-time gameplay during testing.

The last risk is required features and testing taking longer than expected. Some requirements taking longer to implement could consume time to the point where it puts D3 deadlines at risk. Possible mitigation is to prioritizing rubric requirements over features. Optional features such as FR9 will be dropped first if needed, followed by non-essential gameplay polish.

9 Design Depth Profile

DP	Requirement	System Design
DP1	Domain model	Song/Chart module: <code>Song</code> , <code>Chart</code> , <code>Note</code> , <code>TapNote</code> , and <code>HoldNote</code> . Gameplay module: <code>GameSession</code> , <code>PlayerInput</code> , <code>NoteJudge</code> , <code>Score</code> , <code>Combo</code> , and the game-state classes. Progression module: <code>Profile</code> , <code>SongResult</code> , and <code>Progression</code> .
DP2	Variation Points	<code>NoteJudgement</code> , implemented by <code>NormalJudgement</code> <code>ScoringPolicy</code> , implemented by <code>Scoring</code> <code>GameState</code> , implemented by different states.
DP3	Life Cycle	<code>GameSession</code> has four states: Preparing , Playing , Paused , and Finished . Operations such as <code>start()</code> , <code>pause()</code> , <code>resume()</code> , and <code>finish()</code> are delegated to the current state. Invalid transitions are rejected and leave the session unchanged.

DP	Requirement	System Design
DP4	Business Rules	Five domain rules: (1) a note can only be judged once; (2) timing difference determines the judgement according to the active NoteJudgement policy; (3) a successful note increases combo while a Miss resets the current combo; (4) a HoldNote is successful only when its required hold duration is satisfied; and (5) locked songs or difficulties cannot be selected for gameplay.
DP5	Collaboration	UC1 uses GameSession , Chart , Note , NoteJudge , NoteJudgement , ScoringPolicy , Score , and Combo . UC2 uses GameSession , Chart , Score , Combo , SongResult , Profile , and Progression . UC3 uses SaveStore , Profile , and Progression
DP6	Reactive Behaviour	Gameplay performance changes will be exposed through a notification mechanism. When values such as Score or Combo change, interested presentation components can be notified without the domain classes depending directly on Qt widgets.
DP7	External Boundaries	SaveStore provides the persistence boundary, with JsonSaveStore as the real implementation and a mock implementation for testing. Clock provides the timing boundary, with SystemClock as the real implementation and MockClock for deterministic tests. These dependencies are injected through their abstractions.

DP	Requirement	System Design
DP8	Persistence	Profile persists the player's results, unlocked songs, unlocked difficulties, and player information through SaveStore. JSON representation and file-version details are hidden inside JsonSaveStore. Missing, corrupt, or incompatible save data is detected and handled without destroying valid in-memory state.
DP9	Failure Scenarios	Invalid player input: reject the input and preserve the current state. Illegal lifecycle transition: reject the operation and preserve the current state. Corrupt or missing save: report the problem and use a valid default profile without crashing. Invalid chart: reject the chart and prevent the session from starting. Locked content selected: reject the selection and keep the player on the selection screen.
DP10	UI	The Qt interface will provide at least three views: Song Selection, Gameplay, and Results. Song Selection handles song, difficulty, and mode choices. Gameplay displays notes and performance; and Results displays score, accuracy, combo, and progression. UI classes communicate with the core through controllers or interfaces and contain no gameplay logic.

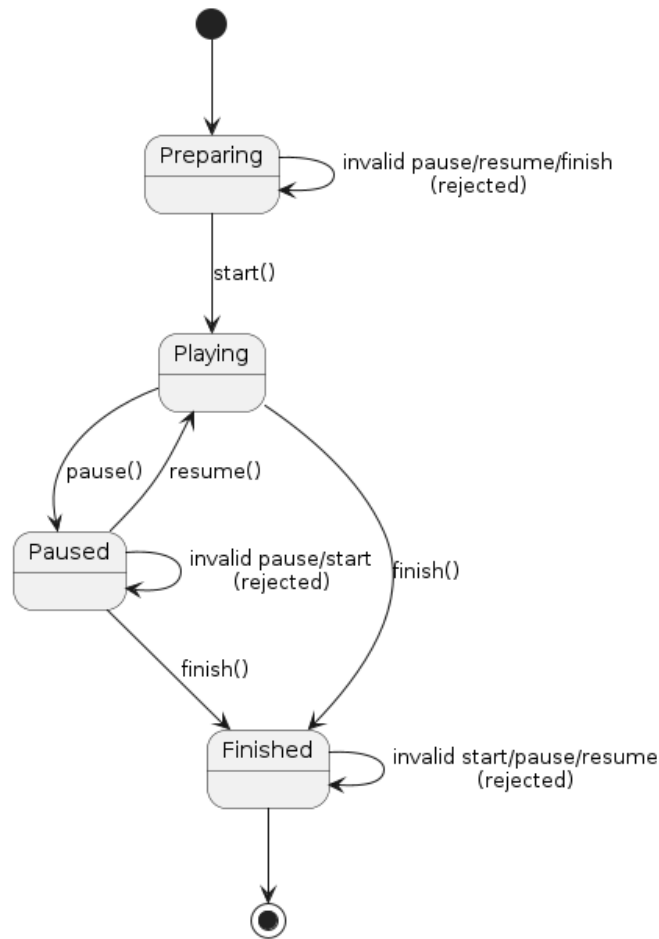


Figure 4: State Diagram

Now, the important lifecycle invariant is that a session cannot return to **Preparing** or **Playing** after reaching **Finished**. And gameplay can't be paused or resumed unless the current state permits that operation. The concrete state classes enforce these transition rules rather than relying on the Qt interface to do so.

R1-R6 will be met by implementing the system using C++ with Qt, and CMake configuration, using 2+ creational patterns and 1 structural pattern with documented design justifications, applying object-oriented approaches, SOLID principles, and low coupling, keeping domain logic in `core/` and Qt portion in `ui/`, validating inputs, rejecting invalid operations, and providing GoogleTest for at least 3 boundaries.